

Inflammatory Disorders of the Nasal Cavities and Sinuses

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 Adapted from: [Inflammatory Disorders of the Nasal Cavities and Sinuses](#)

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Learning Objectives (5)

Versions

After completing this brick, you will be able to:

- 1 Differentiate the use of the terms rhinitis, sinusitis, and rhinosinusitis based on location and causes.
- 2 Differentiate the clinical presentation of rhinosinusitis.
- 3 Explain the pathophysiology of rhinosinusitis.
- 4 Interpret clinical findings to properly diagnose rhinosinusitis.
- 5 Outline the management plan of rhinosinusitis.



CASE CONNECTION

JH is a 50-year-old male with a persistent runny nose and nasal congestion. JH explains, “I am always coughing and clearing my throat. I’ve been doing it for so long, it seems normal. I’m just clearing the drip from my nose.” His symptoms are worse in the spring and late summer and got worse when he moved to a home surrounded by lots of flowering plants and trees. His cough is nonproductive, but he is otherwise well. On exam, the nasal mucosa is edematous with a slightly bluish-pale color. There is a moderate amount of clear, watery nasal discharge. There is no sinus tenderness, and the remainder of his exam is normal.

What will help JH’s symptoms? Consider your answer as you read, and we’ll revisit JH at the end of the brick.

[GO TO CONCLUSION](#) ↓

What Are Rhinitis and Sinusitis (LO #1)?

Both the nose and sinuses are lined by mucosal tissue that is thick with mucous-secreting glands, serving to protect the upper respiratory system from infection by trapping bacteria, preventing them from entering the lungs.

But what happens when the mucosa becomes too zealous in its protective function or becomes infected? That may lead to rhinitis and sinusitis, common inflammatory disorders that are the focus of this brick.

Terminology

Rhinitis is inflammation of the nasal mucosa, which can be allergic or nonallergic:

- **Allergic rhinitis:** what most people commonly refer to as “hay fever” can more properly be called allergic rhinitis due to a type 1 hypersensitivity reaction.
- **Nonallergic rhinitis:** there are many causes for nonallergic rhinitis, including infections (especially viruses), perfumes, odors, chemicals, air pollutants, and changes in temperature.

Allergic and nonallergic rhinitis are contrasted in [Figure 1](#).

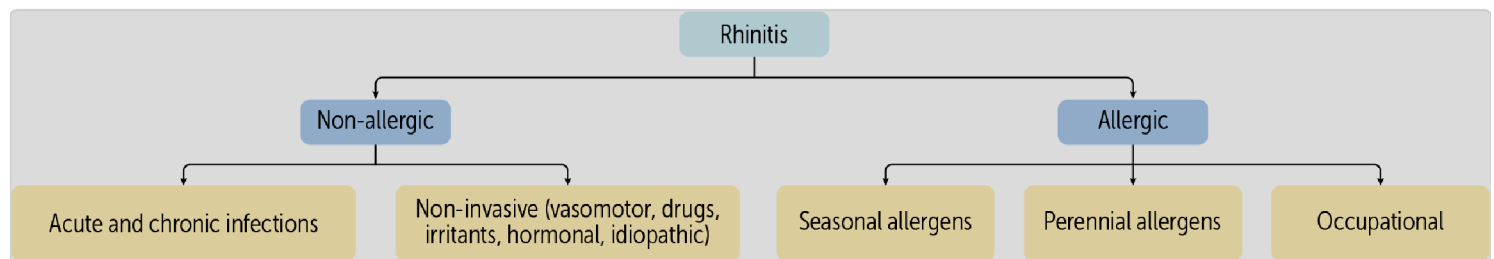


Figure 1 Causes of nonallergic and allergic rhinitis

Sinusitis is inflammation of the mucosa of the paranasal sinuses. It can be caused by an infection (most commonly viral) but also nasal polyps (growths in the sinuses) or a deviated septum (one side of the nose becomes more congested than the other).

If both the paranasal sinuses and the nasal cavity are inflamed simultaneously, as is most often the case, then the condition is referred to as **rhinosinusitis**. Because the conditions often overlap, we'll use the term rhinosinusitis most often here.

Rhinosinusitis can be classified by the time of onset or duration of symptoms:

- **Acute:** symptoms are present for less than 4 weeks.
- **Subacute:** symptoms are present for 4-12 weeks.
- **Chronic:** symptoms have persisted for longer than 12 weeks, usually noninfectious.
- **Recurrent:** >4 episodes/year, lasting 7-10 days with complete resolution of symptoms between episodes

Epidemiology

Rhinosinusitis affects 1 out of every 7 adults in the United States, with more than 30 million individuals diagnosed each year. In the northern hemisphere, rhinosinusitis occurs more often from early fall to early spring since this is when allergens (antigens that commonly induce allergies) peak.

early fall and early spring, when allergens peak.

Rhinosinusitis is most common in the northern hemisphere between

**CLINICAL CORRELATION**

Risk factors for acute rhinosinusitis include older age, smoking, air travel, exposure to changes in atmospheric pressure (e.g., deep sea diving), swimming, asthma and allergies, dental disease, and immunodeficiency.

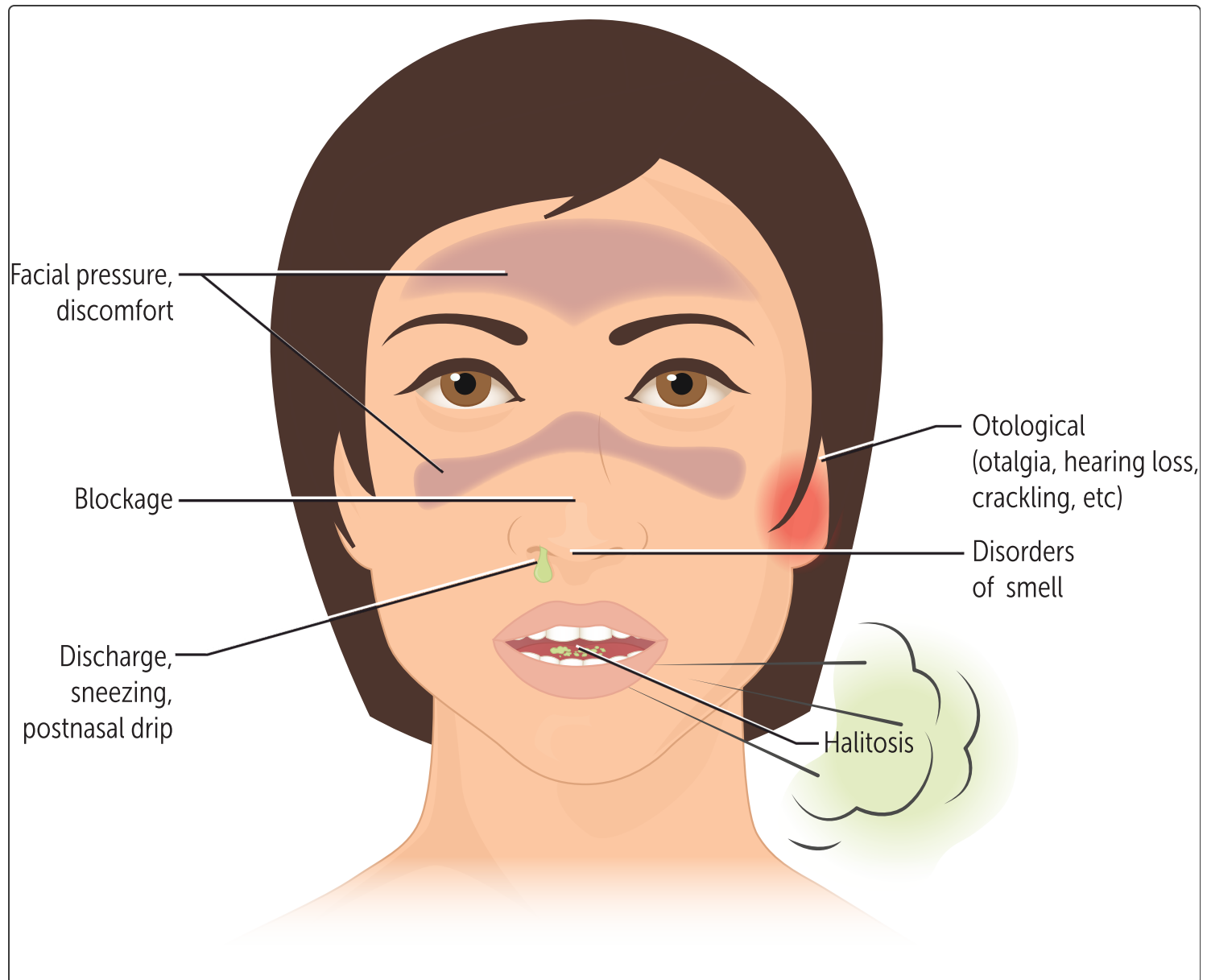
How Do Patients With Rhinosinusitis Present (LO #2)?

Because of the communication between the oral cavity, nasal cavity, and paranasal sinuses, patients with rhinitis and sinusitis experience a variety of overlapping symptoms, including nasal congestion, nasal discharge, postnasal drip (drainage from the nose irritating the back of the throat), nasal itching, sneezing, cough, and fatigue. There may be otalgia (ear pain), anosmia (lack of smell), and halitosis (bad breath).

**CLINICAL CORRELATION**

The nasal discharge in viral rhinosinusitis is typically clear, while bacterial and fungal rhinosinusitis has purulent nasal discharge (thick, yellow-green).

Sinusitis may present with facial pressure or pain. Patients with allergic sinusitis may have other allergy-related symptoms like watery eyes and scratchy throat. The symptoms of rhinosinusitis are summarized in Figure 2.



QUIZ

 Tap image for quiz

Figure 2 Symptoms of sinusitis and rhinitis

Facial fullness or pain is most often present in sinusitis.

What Is the Pathophysiology of Rhinosinusitis (LO #3)?

Infections, allergic responses, and structural causes are most often the causes of rhinosinusitis.

Infections

Infectious rhinosinusitis usually occurs after an upper respiratory infection (common cold). Infections are most commonly viral, but there are also bacterial and fungal causes.

Viruses. The most common viral cause is rhinovirus. Other viruses include adenovirus, respiratory syncytial virus, influenza virus, and parainfluenza virus. Transmission occurs via droplets, aerosols, and contaminated surfaces. Viral infection activates innate immune responses and can impair mucociliary clearance and epithelial integrity, predisposing to secondary bacterial infection.

Bacteria. Bacterial infections usually occur in the sinuses and are much less common than viral infections, causing 1% to 2% of cases. Most are due to *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis*. They usually occur after a viral sinusitis infection due to the altered, obstructed local environment. Bacterial sinusitis is typically suspected when:

- There is an acute, serious onset of symptoms
- Patients have symptoms for more than 10 days without improvement
- There is an initial period of improvement, but the patient's condition then worsens

Fungi. Fungal rhinosinusitis is even less common but can be serious, particularly in its invasive forms. Patients who are immunocompromised (eg, due to HIV infection, transplantation, or chemotherapy) and patients with poorly controlled diabetes mellitus are more susceptible to fungal rhinosinusitis. Organisms include *Aspergillus*, *Fusarium*, and *Mucorales* (*Rhizopus* and *Mucor*) species.

An invasive fungal infection may spread rapidly beyond the paranasal sinuses and nasal cavity into the orbits and brain, causing rhino-orbital-cerebral mycosis. These invasive fungal infections can cause nasal ulcers, facial swelling, and weakness of eye muscles (eg, double vision). It is sometimes fatal.

Complications of Bacterial and Fungal Sinusitis. Severe complications can include extension into the orbits and brain, leading to headaches, neck stiffness, periorbital edema or eye bulging (proptosis) (Figure 3), vision changes, eye pain, or altered mental states. Such patients need emergent management, including CT scanning or MRI to evaluate the potential spread of the infection.



Figure 3. Orbital Cellulitis in a Child.

 CLINICAL CORRELATION

Distinguishing preseptal cellulitis from orbital cellulitis, a severe infection involving the soft tissue behind the orbital septum that requires urgent evaluation and treatment, is important. Patients with orbital cellulitis have ocular pain and eyelid swelling and erythema but may also have pain with eye movements, proptosis, and diplopia suggesting involvement of the orbital tissue.

Preseptal cellulitis is infection of the eyelid skin anterior to the orbital septum, presenting with redness and swelling but no pain with eye movements.

Allergens

Common seasonal allergens that cause rhinosinusitis include grasses, pollen, and weeds. Perennial (nonseasonal) allergens are allergens that are encountered all year long, for example, animal dander, feathers, house dust mites, milk, eggs, nuts, and wheat.

When patients are exposed to allergens, the body recognizes the allergen as an antigen causing a local inflammatory process in the nasal mucosa. **Type 1 hypersensitivity** occurs when an antigen binds to at least two IgE molecules bound to FcεRI receptors on a mast cell or basophil. Cross-linking (close proximity) of the receptors causes the mast cell or basophil to degranulate and release histamine. The histamine leads to local inflammation (Figure 4).

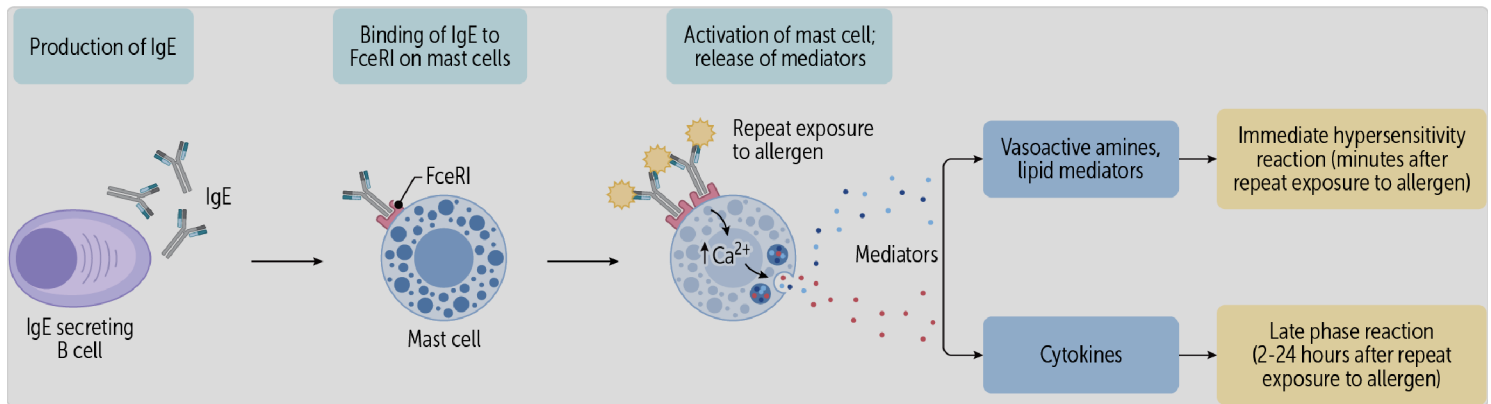


Figure 4 Type 1 hypersensitivity

Structural Causes

Less commonly, rhinitis can be caused by a physical obstruction in the nasal cavity. The most common of these are **nasal polyps**, which are masses composed of swollen mucosa filled with neutrophilic and/or eosinophilic inflammation (Figure 5). The inset shows a CT scan that reveals the mass that is causing a partial obstruction.

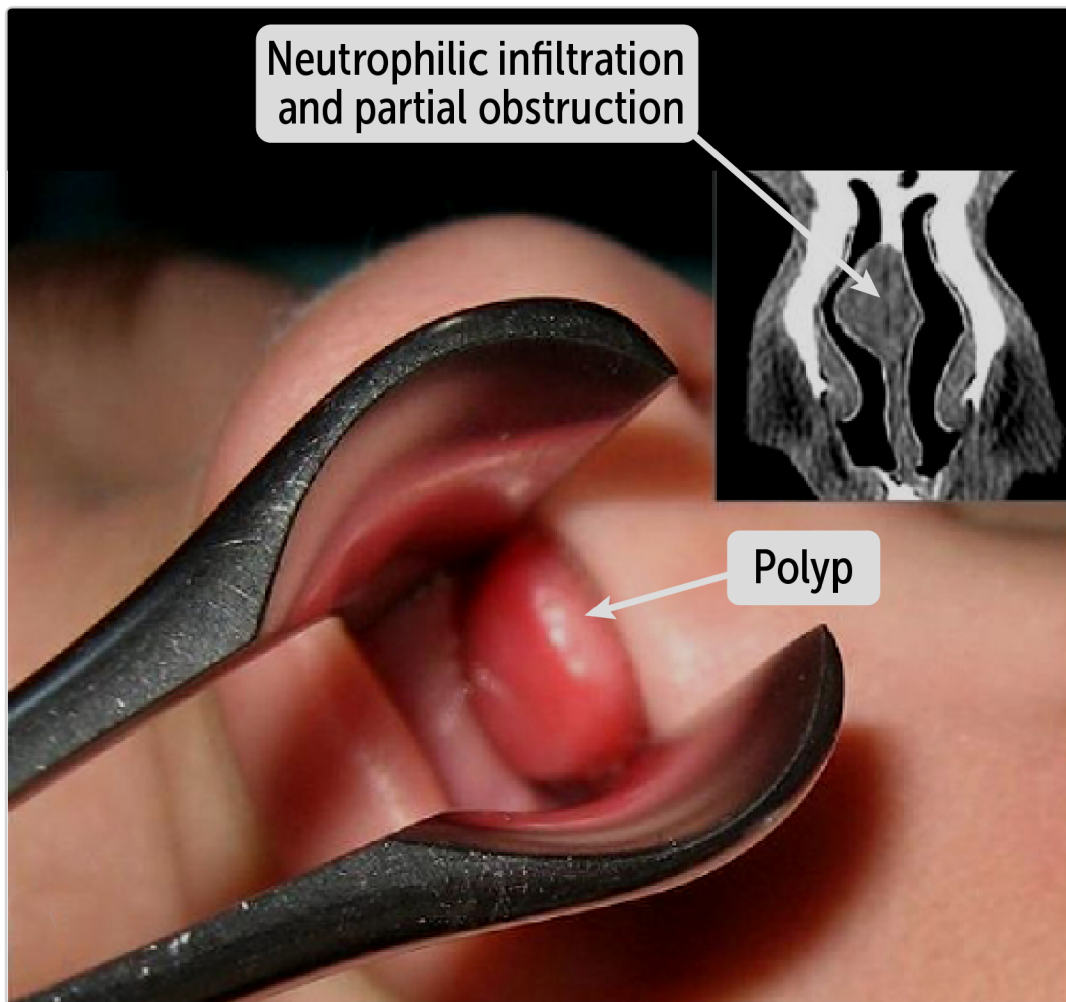


Figure 5 Nasal polyp with CT scan inset

The exact cause of nasal polyps is unknown, but they can be seen as a reaction to aspirin use in some patients.

Other Irritants

Several other factors can cause inflammation of the nasal and sinus mucosae, leading to edema and congestion by nonallergic mechanisms. These include:

- Cold temperature
- Low humidity (dry air)
- Perfumes, odors, chemicals, and air pollutants



CLINICAL CORRELATION

Nasal decongestant sprays can cause a disorder called rhinitis medicamentosa. Many days of regular use of over-the-counter decongestant nasal sprays leads to rebound nasal congestion as the medication wears off, prompting patients to administer the medicine more frequently to obtain relief. This begins a vicious cycle of nasal congestion, both caused and temporarily relieved by the medication, with escalating use and eventual dependency. The risk of developing rhinitis medicamentosa from over-the-counter nasal decongestants can be minimized by limiting use to a maximum of five days, not exceeding recommended frequencies, and using as few doses as possible during those days. There is also evidence that this is less likely to occur if the patient is administering an intranasal corticosteroid along with the nasal decongestant.

Type I hypersensitivity is most common in allergic sinusitis.

How Do We Diagnose Rhinosinusitis (LO #4)?

History and physical exams are usually adequate to diagnose rhinosinusitis, with imaging used in uncertain or severe cases.

Physical Examination

Rhinosinusitis is most often a clinical diagnosis based on history.

Localization of suspected sinusitis can be done by pressing over the facial area. Note that neither palpation nor percussion is sensitive or specific for identifying sinusitis. The physical exam is important in excluding complications like eye findings, neck stiffness, or altered mental state.

Anterior rhinoscopy can be done to look for nasal polyps. Examination of the anterior nasal mucosa with an otoscope may also be useful, as patients with allergic rhinitis can show pale and edematous nasal mucosa, while patients with acute viral rhinosinusitis can show a beefy red nasal mucosa.



CLINICAL CORRELATION

Children with allergic rhinitis may display "allergic shiners" and "adenoid facies," resulting from longstanding mouth-breathing. The features of adenoid facies

include a long, narrow face; narrow, upturned nose; short upper lip; exposed upper incisors; recessed lower jaw; and forward head posture (Figure 6).

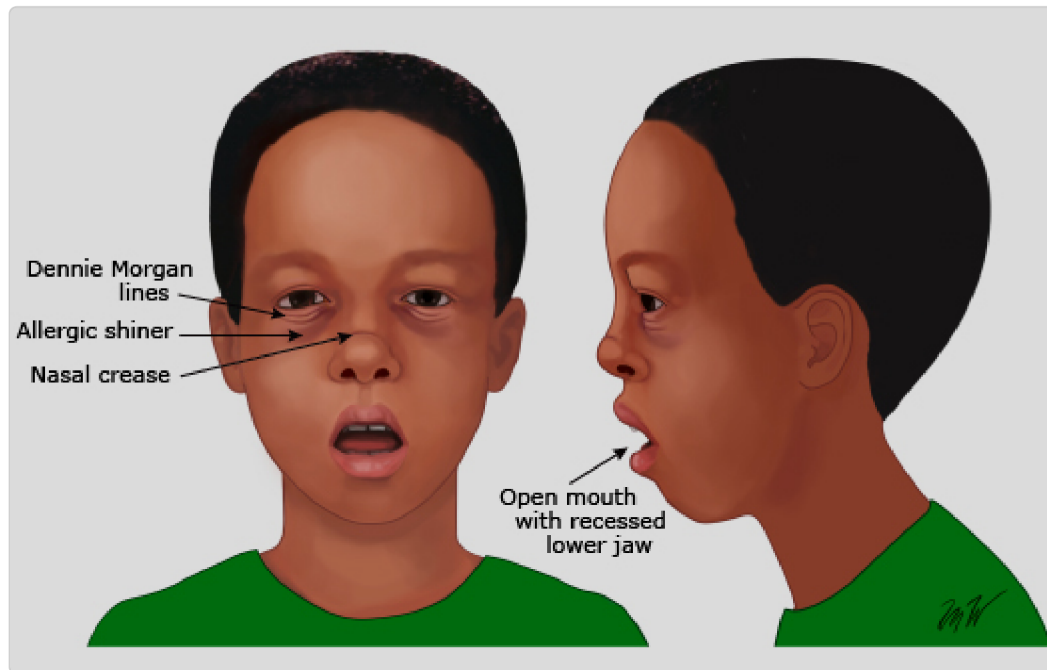


Figure 6. Chronic allergic rhinitis in children.

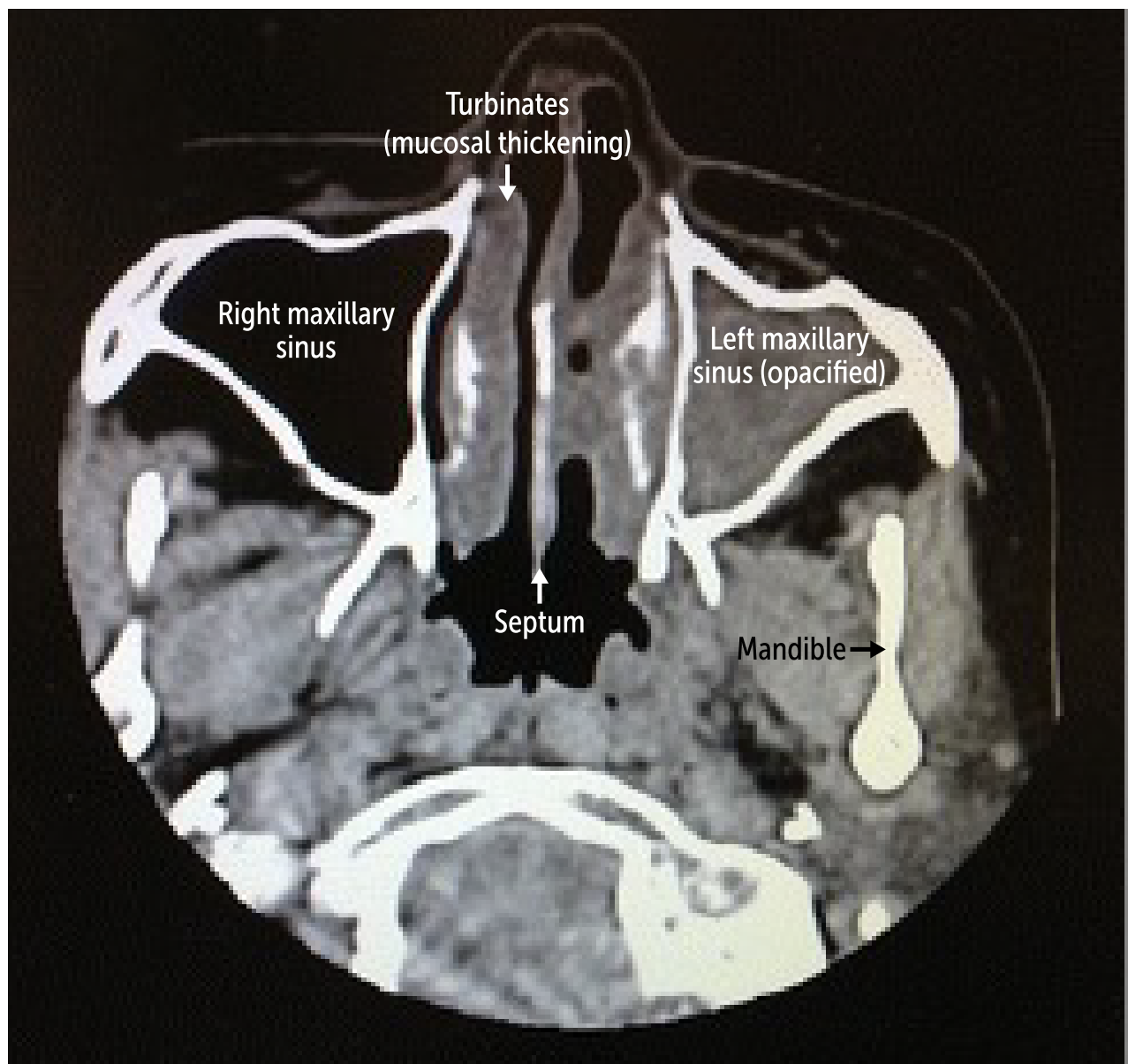
Diagnostic Imaging

Imaging is not done routinely for rhinosinusitis. Indications include one of the complications listed above or suspected chronic sinusitis (>12 weeks of symptoms).

Imaging can locate the spread of bacterial/fungal infections and can also document the extent of sinusitis, including the presence of nasal polyps that may cause it. CT scanning and MRI are both used.

Typical findings seen in chronic sinusitis include mucosal thickening, sinus air-fluid levels, and sinus opacification, which occurs when the sinus is

completely filled with fluid (Figure 7).



QUIZ

👉 Tap image for quiz

Figure 7 Computed tomography image of sinuses with opacification

scanning.

Sinus opacification is a typical sign of severe chronic sinusitis on CT

How Do We Treat Rhinosinusitis (LO #5)?

The treatment varies between infectious and allergic causes.

Viral

Most patients who lack allergic symptoms are presumed to have a viral infection and are initially treated conservatively with symptomatic treatment. Nonsteroidal anti-inflammatory drugs (NSAIDs), acetaminophen, and intranasal saline irrigation are useful.

Additional symptom relief may be achieved with intranasal corticosteroid sprays, such as fluticasone, budesonide, mometasone, triamcinolone, beclomethasone, and ciclesonide. These agents provide local anti-inflammatory effects by binding intracellular glucocorticoid receptors and downregulating inflammatory cytokines, resulting in reduced nasal mucosal edema, improved sinus drainage, and decreased congestion and rhinorrhea. Although onset is not immediate, many patients experience meaningful

improvement within several hours to a few days, with maximal benefit after consistent daily use. Common adverse effects include nasal irritation/burning, dryness, epistaxis, and (less commonly) headache or sore throat. Given the route of administration (i.e., intranasal) it avoids systemic absorption/ effects, but prolonged or high-dose use may rarely contribute to ocular effects (e.g., cataracts/glaucoma) or HPA-axis suppression, particularly in pediatric patients.

For short-term congestion relief, some patients may benefit from topical nasal decongestants such as oxymetazoline (or phenylephrine). Remember these drugs are as α -adrenergic agonists, producing vasoconstriction of nasal mucosal blood vessels and rapid reduction in swelling and congestion. Because of the risk of rebound congestion (rhinitis medicamentosa), intranasal decongestants should be limited to no more than 3-5 consecutive days. Adverse effects may include nasal dryness/irritation, and when overused or systemically absorbed (which is rare), hypertension, tachycardia, palpitations, or insomnia.

	Corticosteroids	α -adrenergic agonists
Common Drugs	fluticasone budesonide mometasone triamcinolone beclomethasone ciclesonide	Oxymetazoline Phenylephrine
Primary Role	Reduces inflammation to improve congestion, drainage, and overall nasal symptoms	Quick relief of nasal congestion by constricting swollen nasal blood vessels

	Corticosteroids	α -adrenergic agonists
MOA	Glucocorticoid receptor agonism → ↓ inflammatory mediators and ↓ mucosal edema	α -adrenergic agonism → vasoconstriction of nasal mucosa → ↓ swelling
Onset	Gradual: hours to days (best with consistent daily use)	Very fast: minutes
Duration/ Usage	Can be used daily for days to weeks as needed	Limit to 3-5 days to prevent rebound congestion

Bacterial

After 7 to 10 days of symptoms, patients who do not improve with conservative therapy may be given antibiotics for presumed bacterial infection. The first-line antibiotics, for adults and pediatrics, is **amoxicillin-clavulanate**. This is because it has good activity against *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis*, as well as other Gram (+) anaerobes that may be in the oral/nasal cavity.

Second-line options, levofloxacin, moxifloxacin, doxycycline, or clindamycin, may be considered if there is a true penicillin allergy or other contraindication to first-line therapy.

Fungal

If the patient is suspected or known to have a fungal infection (e.g., is immunosuppressed), antifungal drugs can be used, and, occasionally, surgical debridement is needed to gain source control.

The antifungal drug is chosen based on suspected organism. See the table below for first-line options.

Table

Causative Organism	First-line Treatment
<i>Aspergillus</i>	Voriconazole (better activity over other -azoles)
<i>Fusarium</i>	Voriconazole or Posaconazole
<i>Mucorales</i> (<i>Rhizopus</i> and <i>Mucor</i>)	Amphotericin B

Allergies

The most effective overall treatment for allergic rhinitis is to avoid the problematic allergen. This may not always be possible, which is where medications can help. Options include:

- Intranasal corticosteroids, as mentioned above, which are first-line therapy.
- Oral or intranasal antihistamines (usually second-generation, e.g., cetirizine, loratadine, fexofenadine).
- Cromolyn is a mast cell stabilizer that prevents histamine release when patients are exposed to allergens. This is rarely used given the better efficacy, less frequent dosing (3-4x/day vs 1-2x/day), and slower onset of action (4-7 days vs <1 day) of corticosteroids and antihistamines.
- Intranasal saline, i.e., saline wash, can be helpful for some patients to clear the nasal passage.

Intranasal corticosteroids are the first-line therapy for allergic rhinitis.



CASE CONNECTION

[BACK TO INTRODUCTION](#) ↑

Thinking back to JH, what will help his symptoms?

You explain to JH that he has allergic rhinitis, sometimes called hay fever. You explain, “It sounds like your symptoms are worse when there is a lot of pollen in the air; it can be difficult to avoid pollen if you live in a place that is surrounded by plants. I can prescribe a nasal spray, an antihistamine called azelastine, that will reduce your allergy symptoms and should not have side effects such as dry mouth or sleepiness.” JH agrees to this plan and to follow up if he does not improve. You receive a message from JH a week later, letting you know that the medication started working right away.

Summary

- Rhinitis, an inflammation of the nasal canal, can be allergic or nonallergic (ie, due to infections).
- Nonallergic causes include infections and irritants.
- Sinusitis is an inflammation or infection of the paranasal sinuses, often concurrent with rhinitis (and termed rhinosinusitis).

- Rhinosinusitis can be divided into acute, subacute, chronic, and recurrent based on the length of time the patient has symptoms.
- Symptoms include nasal congestion, nasal discharge, facial pain, facial pressure, ear pain, halitosis, headaches, and anosmia.
- Viral infections are the most common cause.
- Bacterial infections are mainly caused by three organisms: *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis*.
- Fungal infections are usually noninvasive but can spread rapidly in immunocompromised patients and invade structures outside the sinuses and nasal cavities.
- Allergic rhinitis is a type 1 hypersensitivity reaction and occurs after exposure to allergens.
- Nasal polyps may obstruct the nose and cause rhinitis.
- Diagnosis is initially clinical.
- Imaging is indicated for chronic sinusitis or severe symptoms.
- Acute viral sinusitis is treated initially with nonsteroidal anti-inflammatory drugs (NSAIDs), acetaminophen, intranasal saline irrigation, and intranasal corticosteroids.
- Acute bacterial sinusitis is treated with antibiotics.
- Patients with allergic rhinitis are treated with intranasal steroids and intranasal antihistamines.

Review Questions

1. Which of the following findings on a CT scan are most suggestive of chronic sinusitis?

A. Fracture of the maxillary bone

- B. Mass posterior to the sphenoid sinus
- C. Mucosal thickening of the frontal sinus
- D. Protuberance of occipital bone
- E. Ring enhanced lesion of temporal lobe

Explanation (requires correct answer)

2. Which of the following organisms is the most likely cause of acute bacterial sinusitis?

- A. *Klebsiella pneumoniae*
- B. *Legionella pneumophila*
- C. *Mycoplasma pneumoniae*

- D. *Staphylococcus aureus*
- E. *Streptococcus pneumoniae*

Explanation (requires correct answer)

3. A 12-year-old male has been experiencing clear, watery nasal discharge since the beginning of April and is found to have an allergy to ragweed. Which of the following types of hypersensitivity reaction is most associated with his likely condition?

- A. Type 1 hypersensitivity reaction
- B. Type 2 hypersensitivity reaction
- C. Type 3 hypersensitivity reaction
- D. Type 4 hypersensitivity reaction

Explanation (requires correct answer)

4. Which of the following is the best initial treatment for acute viral sinusitis?

- A. Amoxicillin-clavulanate
- B. Clindamycin
- C. Cromolyn
- D. Intranasal antihistamine
- E. Intranasal saline spray

Explanation (requires correct answer)

5. What is the term for simultaneous inflammation of the paranasal sinuses and nasal cavity?

- A. Sinusitis
- B. Rhinosinusitis
- C. Nasal polyps
- D. Deviated septum
- E. Rhinorrhea

Explanation (requires correct answer)



Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 Differentiate the use of the terms rhinitis, sinusitis, and rhinosinusitis based on location and causes.
- 02 Differentiate the clinical presentation of rhinosinusitis.
- 03 Explain the pathophysiology of rhinosinusitis.
- 04 Interpret clinical findings to properly diagnose rhinosinusitis.
- 05 Outline the management plan of rhinosinusitis.

Objective Confidence: 0/5